# MBS-raw: Physical Optics for Ice Crystals

Manual

2026-03-22

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# 1 Overview

MBS-raw computes light scattering by non-spherical particles using the Physical Optics (PO) method. It traces geometric optics rays through the particle (reflections, refractions), then applies Kirchhoff diffraction to each output beam to obtain the far-field Mueller matrix.

**Optimized version**: ∼70× faster than original. AVX-512/AVX2 SIMD, OpenMP parallel, Sobol quasi-random orientations. Builds for Intel, AMD Zen 2 (EPYC 7H12), and AMD Zen 4 (Ryzen 7000 / EPYC Genoa).

# 2 Quick Start

```
# Simple run: hex column, Sobol orientations, auto theta grid
mbs_po --po --sobol 1024 --auto_tgrid \
    -p 1 10 10 -w 0.532 --ri 1.31 0 -n 12 \
    --grid 0 180 48 180 --close

# Fully adaptive (auto everything)
mbs_po --po --adaptive 0.01 --auto_tgrid \
    -p 1 10 10 -w 0.532 --ri 1.31 0 -n 12 \
    --grid 0 180 48 180 --close
```

# 3 CLI Reference

All flags are grouped by function. Required flags have no default; optional flags are marked with their defaults.

## 3.1 Particle Definition

### 3.1.1 `-p TYPE HEIGHT DIAMETER [EXTRA]`

| Type | Shape | Parameters |
|---|---|---|
| 1 | Hexagonal column | H = height, D = diameter |
| 2 | Bullet | H, D (cap angle 62°, auto) |
| 3 | Bullet rosette | H, D, [cap height] |
| 4 | Droxtal | arg3 = sup parameter |
| 10 | Concave hexagonal | H, D, concavity depth |
| 12 | Hexagonal aggregate | H, D, number of elements |

**Symmetry**: auto-detected. Hex column: $\beta_{\text{sym}} = \pi/2$, $\gamma_{\text{sym}} = \pi/3 \rightarrow 12\times$ reduction.

### 3.1.2 `-pf FILENAME`

Load particle geometry from file. `-p` argument provides the filename.

### 3.1.3 `-rs SIZE`

Resize particle loaded with `-pf` to maximal dimension SIZE ($\mu$m).

### 3.1.4 `-forced_convex` / `-forced_nonconvex`

Force convex or non-convex tracing algorithm.

## 3.2 Physical Parameters

### 3.2.1 `-w WAVELENGTH_UM`

Wavelength in micrometers. Examples: `-w 0.532` (green), `-w 1.064` (NIR).

### 3.2.2 `-ri N_REAL N_IMAG`

Complex refractive index $m = n_{\text{re}} + i\, n_{\text{im}}$.
**Important**: Without `-abs`, imaginary part is forced to zero.

### 3.2.3 `-abs`

Enable absorption. Requires `-w`. Uses `DiffractInclineAbs`.

### 3.2.4 `-n MAX_REFLECTIONS`

| Value | Use case | Notes |
|---|---|---|
| 5–6 | Forward scattering, $C_{\text{sca}}$ | $< 1\%$ error |
| 12 | General purpose | $< 5\%$ backscatter error |
| 20 | Backscattering | $< 3\%$ error |
| 25 | Reference calculations | |

## 3.3 Method

- `-po` — Physical Optics (Kirchhoff diffraction). **Required for PO.**
- `-go` — Geometrical Optics (no diffraction).
- `-incoh` — Incoherent summation ($|J|^2$ per beam, then sum).
- `-karczewski` — Karczewski polarization matrix (experimental, same $M_{11}$).
- `-jones` — Output Jones matrices to file.
- `-all` — Compute all trajectory groups.

## 3.4 Orientations

### 3.4.1 `-sobol N` (recommended)

$N$ Sobol quasi-random orientations (power of 2). Auto-uses particle symmetry.
Convergence: $O((\log N)^2/N)$, much faster than Monte Carlo $O(1/\sqrt{N})$.

| $x$ | $C_{\text{sca}}$ ($< 2\%$) | $M_{11}(180°)$ ($< 5\%$) |
|---|---|---|
| 10–50 | 128 | 512 |
| 50–200 | 256 | 1024 |
| 200–1000 | 512 | 4096 |
| $> 1000$ | 1024 | 8192+ |

### 3.4.2 `-adaptive EPS`

Auto-determine orientations. Starts at 256, doubles until $C_{\text{sca}}$ converges to relative accuracy EPS.

### 3.4.3 `-sym B G`

Override particle symmetry: $\beta \in [0, \pi/B]$, $\gamma \in [0, 2\pi/G]$.
Example: `-sym 6 2` $\rightarrow \beta \in [0, 30°]$, $\gamma \in [0, 180°]$ ($12\times$ reduction).

### 3.4.4 Other orientation modes

- `-random N_BETA N_GAMMA` — Uniform grid.
- `-montecarlo N` — Random orientations.
- `-fixed BETA GAMMA` — Single orientation (degrees).
- `-orientfile FILE` — Read $(\beta, \gamma)$ pairs (radians) from file.
- `-b MIN MAX`, `-g MIN MAX` — Override ranges (degrees).

## 3.5 Scattering Grid

### 3.5.1 `-grid T1 T2 N_PHI N_THETA`

Uniform grid. **Use** $N_\phi \geq 48$; $N_\phi = 6$ causes 50–200% error.

### 3.5.2 `-auto_tgrid` (recommended)

Auto non-uniform $\theta$ grid with 5 adaptive zones:

| Zone | Range | Step | Purpose |
|---|---|---|---|
| Fine | 0 to 10×peak | peak/20 | Forward peak |
| Transition | to 20° | geometric ×1.3 | Smooth bridging |
| Medium | to 120° | 0.5° ($x$<100) or 1° | Side scattering |
| Side/back | to 175° | 1° | Backscattering |
| Near-backscatter | to 180° | 0.25° | LDR |

[width=]docs/figures/fig$_a uto_g rid_p oints.pdf$

Figure 1: Left: total $N_\theta$ and $N_\phi$ vs size parameter $x$. Right: zone breakdown of $\theta$-grid points.

[width=0.85]docs/figures/fig$_a uto_g rid_s teps.pdf$

Figure 2: Step size $\Delta\theta$ vs scattering angle for different $x$. Small $x$ has coarser steps (broad peak); large $x$ needs fine steps near $\theta$=0.

### 3.5.3 `-auto_phi`

Auto-select $N_\phi$ based on $x$: $N_\phi = \max(48,\ 6\lceil\sqrt{x}/1.5\rceil)$.

[width=]docs/figures/fig$_a uto_p hi.pdf$

Figure 3: Left: $N_\phi$ from auto_phi (blue) vs fixed values. Right: $Q_{\text{sca}}$ error vs $N_\phi$ for different sizes—larger $x$ needs more $\phi$ bins.

**Recommended**: use both `-auto_tgrid -auto_phi` together.

### 3.5.4 `-tgrid FILENAME`

Custom $\theta$ grid from file (degrees, one per line).

## 3.6 Multi-Size (`-sizefile`)

```
--sizefile sizes.txt   # one x = pi*D/lambda per line
```

Traces beams once, caches geometry, recomputes diffraction for each size. Output: `M_x10.dat`, `M_x20.dat`, etc.

### 3.7 Performance Options

- `-beam_cutoff EPS` — Skip negligible beams.
- `-shadow_off` — Disable shadow beam (debug).
- `-r RATIO` — Small beam restriction ratio (default 100).

### 3.8 Output

- `-o DIRNAME` — Output folder (default `M`).
- `-close` — Exit after calculation (**required for scripts**).
- `-log SECONDS` — Progress output interval.
- `-tr FILE` — Trajectory groups from file.
- `-gr` — Per-group output.

## 4 Output Files and Console Log

### 4.1 Output files

All output is written to the directory specified by `-o` (default `M`).

#### 4.1.1 `DIRNAME.dat` — Mueller matrix

Space-separated, 18 columns, one header line + one row per $\theta$ bin.

```
ScAngle  2pi*dcos  M11  M12  M13  M14  M21  M22  M23  M24
                   M31  M32  M33  M34  M41  M42  M43  M44
```

- `ScAngle` — scattering angle $\theta$ in degrees
- `2pi*dcos` — solid angle weight: $2\pi\,\Delta\cos\theta$. Integration: $C_{\text{sca}} = \sum_j M_{11}(\theta_j) \times (2\pi\,\Delta\cos\theta_j)$
- `M11`... `M44` — 16 Mueller matrix elements (phi-averaged). For randomly-oriented particles: $M_{13} = M_{14} = M_{23} = M_{24} = M_{31} = M_{32} = M_{41} = M_{42} \approx 0$.

Useful derived quantities:

$$
\begin{aligned}
Q_{\text{sca}} &= C_{\text{sca}}/A_{\text{proj}} \\
g &= \frac{\sum M_{11}\cos\theta \cdot 2\pi\, d\cos\theta}{\sum M_{11} \cdot 2\pi\, d\cos\theta} \qquad \text{(asymmetry parameter)} \\
-M_{12}/M_{11} &= \text{degree of linear polarization}
\end{aligned}
$$

#### 4.1.2 `DIRNAME\out.txt` — run summary

Contains particle parameters, refractive index, method, start/end time, total number of orientations. Written after computation completes.

```
Particle:  Refractive index: 1.310000
  Area:429.903838
Number of secondary reflections: 6
Wavelength (um): 0.532000
Method: Physical optics, Sobol quasi-random
Start of calc.: Sun Mar 22 11:22:44 2026
End of calc.  : Sun Mar 22 11:22:49 2026
Total time  : 4s
Total number of body orientations: 128
```

#### 4.1.3 `DIRNAME_jones.dat` — Jones matrices (with `-jones`)

Written when `-jones` flag is used (fixed orientation only). 10 columns:

```
# theta(deg) phi(deg) J00_re J00_im J01_re J01_im J10_re J10_im J11_re J11_im
```

One row per $(\theta, \phi)$ grid point. Complex Jones matrix $\mathbf{J}$ relates incident to scattered field: $\begin{pmatrix} E_\parallel \\ E_\perp \end{pmatrix}_{\text{sca}} = \mathbf{J} \begin{pmatrix} E_\parallel \\ E_\perp \end{pmatrix}_{\text{inc}}$.

#### 4.1.4 `particle_for_check.dat` — particle geometry

Dumped by the program for verification. Contains vertex coordinates and facet definitions of the particle as used in the calculation.

#### 4.1.5 `beam_log.dat` — beam diagnostics

Written for the first orientation (debug). Columns:

```
# beam_num  nActs  area  opticalPath  |Jones|  cx  cy  cz  nVertices
```

`nActs` = number of internal interactions, `cx,cy,cz` = beam exit direction. The shadow beam has `nActs` = large number and `cz` = $-1$.

### 4.2 Console output (stdout)

The program writes progress and results to stdout.

#### 4.2.1 Auto mode output

```
Auto theta grid: x=83.5, peak_width=2.16 deg, 464 points
Auto phi: x=83.5, N_phi=48
Adaptive mode: target relative change = 0.01
  N=64  (+64 new), M11(180)=7.09, dM11=100%, Csca=92.9, dCsca=100%
  N=128 (+128 new), M11(180)=6.69, dM11=5.6%, Csca=93.0, dCsca=0.12%
  N=256 (+256 new), M11(180)=6.64, dM11=0.8%, Csca=92.9, dCsca=0.08%
Converged at N=256 (both C_sca and M11)
```

**Convergence messages**:

- `Converged (both C_sca and M11)` — best: both criteria met
- `Converged (C_sca ok, M11 within X%)` — relaxed: M11 within $5 \times \varepsilon$
- `Converged (C_sca stable, M11 still X%)` — C_sca $< \varepsilon/10$ after 4 iterations
- `WARNING: Max orientations reached` — did not converge, result approximate

#### 4.2.2 Sobol mode output

```
Sobol: 1024 orientations, beta_sym=30 deg, gamma_sym=180 deg
Phase 1 (tracing + preprocessing): 0.46 s
Phase 2 (diffraction, OpenMP): 13.25 s
Total: 13.71 s
```

#### 4.2.3 Scattering efficiency (always printed)

```
===== SCATTERING EFFICIENCY =====
C_sca = 160.0398
A_proj (incoming energy) = 93.0208
Q_sca = C_sca / A_proj = 1.7205
================================
```

If $Q_{\text{sca}} > 2.5$: `WARNING: Q_sca > 2.  PO overestimates at this size.`

### 4.3 Console output (stderr)

#### 4.3.1 Memory diagnostic

```
Memory: 25219 MB available, chunk=4096 orientations (1 chunks)
Adaptive: max orientations = 32768 (25219 MB available)
```

#### 4.3.2 Timing breakdown (at program exit)

```
===== HandlerPO TIMING BREAKDOWN =====
HandleBeams calls: 448
HandleBeams total:    69.255 s
  Beam loop total:    66.039 s
  AddToMueller:       2.606 s
  Unaccounted:        63.433 s
=====================================
```

"Unaccounted" = the optimized inline loop (batched sincos + edge loop + rotate_jones). Not instrumented to avoid timing overhead.

# 5 Build

## 5.1 Intel / AVX-512

```
bash build.sh            # -> bin/mbs_po
```

`-O3 -march=native -mavx512f -mavx512dq -fopenmp`. Requires GCC $\geq 9$.

## 5.2 AMD EPYC 7H12 (Zen 2, SP3)

```
bash build_epyc.sh       # -> bin/mbs_po_epyc
bash build_epyc_clang.sh # -> bin/mbs_po_epyc_clang
```

`-march=znver2`. No AVX-512; `fast_sincos_8x` falls back to 2×`fast_sincos_4x`.

**NUMA**: `OMP_PROC_BIND=close OMP_PLACES=cores OMP_NUM_THREADS=64`

## 5.3 AMD Zen 4 (Ryzen 7000, EPYC Genoa)

```
bash build_zen4.sh       # -> bin/mbs_po_zen4
```

`-march=znver4 -mavx512f -mavx512dq -mavx512vl`.

## 5.4 Generic AVX2

Remove `-mavx512*` from `build.sh`. Code auto-detects via `#ifdef __AVX512F__`.

# 6 Performance

## 6.1 Single-thread benchmarks

Hex $D$=$H$=10 $\mu$m, 128 Sobol, 48 $\phi$ × 181 $\theta$, $n$=6:

| Version | Phase 2 | vs Original |
|---|---|---|
| Original (before optimizations) | ∼320 s | 1× |
| Optimized + batched sincos | 4.6 s | ∼**70**× |
| EPYC build (AVX2 only) | 4.7 s | ∼68× |

## 6.2 Key optimizations

1. **PrecomputeEdgeData**: slopes, intercepts computed once per beam (not per direction).
2. **Theta-coefficients**: $A(\theta) = \sin\theta \cdot a_s + \cos\theta \cdot a_c + a_0$ (2 FMA).
3. **Batched sincos**: all vertex phases for all $\theta$ pre-computed before the $\theta$-loop. AVX-512 `fast_sincos_8x` or AVX2 2×`fast_sincos_4x`.

4. **Polynomial sincos**: Cody–Waite range reduction + Chebyshev polynomial (∼14 decimal digits).
5. **Inline hot loop**: no heap allocation, stack-only `complex` arithmetic.
6. **OpenMP**: thread-local Mueller accumulators, `schedule(dynamic,1)`.
7. **Sobol + symmetry**: native C++ Sobol generator with particle symmetry reduction.

### 6.3 OpenMP scaling

| Threads | Phase 2 | Speedup |
|---|---|---|
| 1 | 4.6 s | 1.0× |
| 4 | 1.7 s | 2.7× |
| 12 (SMT) | ∼0.9 s | ∼5× |
| 64 (EPYC) | ∼0.2 s* | ∼23×* |

*Estimated; use ≥512 orientations for 64 cores.

## 7 Bugfixes

### 7.1 Forward-direction Fresnel sign (2026-03-22)

**Bug**: sign error in the forward-direction special case ($A{\approx}0, B{\approx}0$) of the Kirchhoff integral. The edge-sum converges to $-F_{\text{inv}} \cdot \text{area}$ but the shortcut used $+F_{\text{inv}} \cdot \text{area}$.

**Symptom**: $M_{11}(0°)$ anomalously small in coherent mode for small particles. Example (hex $D{=}H{=}3\,\mu$m): $M_{11}(0)/M_{11}(0.1°) = 0.019$ (should be $\approx 1.03$).

**Fix**: negate the forward-direction formula in 6 locations:

- `HandlerPO.cpp`: lines 970, 1333, 1889
- `Handler.cpp`: lines 255, 473, 642

Incoherent mode unaffected ($|J|^2$ is sign-independent). $Q_{\text{sca}}$ changed by ∼0.06%. See `BUGFIX_forward_direct` for full details.

## 8 Known Limitations

1. $Q_{\text{sca}} > 2$ **at large** $x$: PO does not enforce the optical theorem.
2. $M_{33}, M_{34}, M_{44}$: RotateJones vs Karczewski give different off-diagonal elements (same $M_{11}$).
3. **Small particles** ($x < 20$): PO not valid. Use ADDA or T-matrix.
4. **Backscattering convergence**: $M_{11}(180°)$ needs 10–100× more orientations.
5. **LTO**: `-flto` causes ∼20% regression. Do not use.

## 9 File Listing

| File | Description |
|---|---|
| `build.sh` | Intel/AVX-512 build |
| `build_epyc.sh` | AMD EPYC 7H12 (Zen 2) build |
| `build_epyc_clang.sh` | EPYC + Clang/AOCC build |
| `build_zen4.sh` | AMD Zen 4 (Ryzen 7000/Genoa) build |
| `Makefile` | Legacy Makefile (Intel) |
| `Makefile.epyc` | Makefile for EPYC |
| `tests/run_tests.sh` | Regression test suite |
| `MANUAL.md` | This manual (Markdown source) |
| `BUGFIX_forward_direction_sign.md` | Bugfix documentation |